

Personal Statement

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UK Global Talent Visa — Digital Technology (Exceptional Promise)

I am an independent researcher and systems engineer based in Derby, United Kingdom. With a formal educational background in Computer Science and Engineering, my core specialisations lie at the intersection of Scientific Computing, Computational AI, and Cyber Security. Over the past three years—without institutional affiliation, funding, or supervision—I have designed and built a production-grade blockchain compliance system, developed original machine learning frameworks, and published research spanning financial systems, energy infrastructure, and applied mathematics.

My work focuses on a fundamental challenge: detecting and preventing failure in complex systems before it occurs. I am applying under the Exceptional Promise route as my work demonstrates both advanced technical capability and clear potential to contribute to the UK's fintech and scientific ecosystem.

1. AMTTP: Pre-Settlement Compliance Infrastructure

My core project is the Anti-Money Laundering Transaction Trust Protocol (AMTTP).

Existing compliance systems—both in traditional finance and blockchain—operate after transactions are executed, limiting their ability to prevent risk. AMTTP introduces a pre-settlement model, where compliance decisions are enforced before transaction finality within the Ethereum block window (~12 seconds).

The system performs:

- Machine learning risk scoring
- Graph-based relationship analysis
- Sanctions screening
- Policy enforcement

before allowing transactions to proceed.

Architecture Overview

AMTTP is implemented as a four-layer system:

- **Client Layer:** Open-source SDKs (TypeScript and Python) with 17–19 service modules (risk, KYC, transactions, disputes, sanctions, monitoring).
- **Compliance Layer:** 7 parallel microservices (ML engine, graph analysis, policy engine, sanctions, monitoring, geographic risk, explainability).
- **ML Pipeline:** Trained on 2.64 million Ethereum transactions, achieving ROC-AUC 0.9879.

- **Blockchain Layer:** 18 Solidity smart contracts enforcing decisions (Approve, Review, Escrow, Block).

The system runs across 17 containerised microservices, with API gateway security, cryptographic signing, rate limiting, and monitoring.

I also developed:

- A cross-platform Flutter application (6 platforms)
- A Next.js compliance dashboard (“War Room”)
- Wallet integrations for real-time user interaction

2. Privacy-Preserving Compliance (zkNAF)

To address the conflict between regulatory transparency and user privacy, I developed zkNAF, a zero-knowledge compliance framework.

It enables:

- Sanctions verification without revealing identities
- Risk validation without exposing scores
- KYC confirmation without disclosing personal data

This allows compliance enforcement while maintaining alignment with GDPR data minimisation principles.

3. Research: Blind Spot Decomposition Theory (BSDT)

Alongside the engineering work, I developed Blind Spot Decomposition Theory (BSDT).

BSDT addresses a key limitation in machine learning systems: the inability to detect risks arising from hidden or unobserved factors. It provides a structured framework for identifying these “blind spots” and improving detection performance without retraining models.

This work has been validated across multiple datasets and domains, forming the analytical foundation of the AMTTP risk engine.

4. Cross-Domain System Stability

Building on BSDT, I extended the framework into a broader approach for analysing system instability across domains.

The same underlying principles can be applied to:

- Financial system risk detection
- Blockchain transaction analysis
- Infrastructure instability (e.g. energy systems)

- Complex dynamic systems

This demonstrates a unified approach to failure detection, with potential applications beyond fintech.

5. Intellectual Property and Open Source

I have filed a UK patent application (filed 4 March 2026, comprising 25 claims) covering my work across:

- Machine learning risk modelling
- Blockchain-based compliance systems
- Cross-domain stability analysis

I have also open-sourced key components of my work, including:

- TypeScript and Python SDKs
- Machine learning pipelines
- Smart contract implementations
- Systemic risk modelling tools

6. Commercial Strategy & Market Potential

The commercial gap AMTTP fills is significant. Current on-chain compliance providers (e.g., Chainalysis, Elliptic, TRM Labs) operate almost exclusively post-settlement, which exposes institutions to irreversible operational risk once funds have moved. AMTTP provides the missing infrastructure for pre-trade compliance—an absolute requirement for regulated financial institutions and Virtual Asset Service Providers (VASPs) entering decentralised finance (DeFi).

From a scalability and commercial readiness perspective:

- The system architecture successfully supports batch-scoring up to 10,000 transactions simultaneously.
- The smart contract layer is actively deployed and validated on the Ethereum Sepolia testnet.
- The design explicitly caters to FCA and FATF Travel Rule obligations, enabling fluid compliance without compromising GDPR principles.

By providing a “Compliance-as-a-Service” (CaaS) API that computes verdicts within the ~12-second block confirmation window, AMTTP holds clear commercialisation potential within the B2B enterprise fintech sector. My UK patent establishes the defensive moat necessary to commercialise this product.

7. Why the UK

The UK is a global leader in financial regulation, fintech innovation, and applied mathematics.

My work aligns directly with national priorities:

- **FCA:** regulatory compliance for digital assets
- **Bank of England / PRA:** systemic risk and financial stability
- **Research ecosystem:** machine learning and applied mathematics

My immediate plans are to:

- Publish my research in peer-reviewed venues
- Develop AMTTP into a compliance platform for regulated institutions
- Build a UK-based company at the intersection of machine learning, fintech, and system stability

8. Closing Statement

Working independently over the past three years, I have:

- Built a full compliance protocol with 17 microservices and 18 smart contracts
- Trained ML systems on 2.64 million blockchain transactions
- Developed original research spanning multiple domains
- Filed a UK patent application
- Open-sourced the system

What distinguishes my work is the ability to connect deep analytical thinking with production-grade systems, and to apply these ideas across different domains.

I believe I have the potential to contribute meaningfully to the UK's digital economy and research landscape, and I am seeking the opportunity to continue this work as a computer scientist in the United Kingdom.